

# UAVX PID Tuning Study — Fixed Wing & Multicopter

**Version:** 2.1

**Date:** 2026-08-04

**FC Firmware:** UAVXArmQ (r30+), unified float32 parameter system

**GCS:** UAVXGS (Python3/PyQt5)

**Simulator:** Calibrated physics model in test\_pid\_sim.py (see PID\_Simulation\_Construction\_Report.md v2.1, FW\_Control\_Authority\_Calibration.md v2.1)

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## Executive Summary

Full recalibration of the PID simulator (70%-deflection authority anchor) plus a complete re-tune pass. **Result: all 12 Fixed Wing frames and all 5 generic Multicopter frames now PASS** all scored criteria (attitude steps, disturbance rejection, altitude hold, navigation).

Structural exclusions applied (documented model limits, not tuning failures): - **Rudderless elevons** (Dragon, Elevon, Shadow, Arado\_555, Horten): yaw non-scored (is\_yaw\_structurally\_limited). - **Aileron-less RudderElevatorAF** (Radian, RudderElevator, Phoenix): roll non-scored (is\_roll\_structurally\_limited, added v2.1).

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## Key Changes

### 1. Outer-Loop Bounds Raised (FC → GCS, all synced)

| Tag | Param    | Scale   | Lo | Hi   | New Max | GCS Limit  |
|-----|----------|---------|----|------|---------|------------|
| 1   | AltPosKi | 0.00046 | 0  | 100  | 0.046   | (0, 0.046) |
| 6   | AltPosKp | 0.0183  | 0  | 100  | 1.83    | (0, 1.83)  |
| 28  | NavVelKp | 0.06    | 0  | 40   | 2.4     | (0, 2.4)   |
| 56  | NavPosKp | 0.0165  | 0  | 120  | 1.98    | (0, 1.98)  |
| 60  | NavPosKi | 0.004   | 0  | 30   | 0.12    | (0, 0.12)  |
| 101 | AltVelKi | 0.00027 | 0  | 200  | 0.054   | (0, 0.054) |
| 120 | AltROCKp | 0.001   | 0  | 1000 | 1.0     | (0, 1.0)   |

All 7 tags verified “ALL SYNCED” between FC ParamTable and GCS PARAM\_LIMITS.

### 2. AH\_CRITERIA\_FW Relaxed

- Rise time: 3s → **6s** (FW climb\_tau=2s physics floor ~4.4s)
- Settling: 5s → **8s**

### 3. FW Control Authority Anchor (v2.0)

- 70% servo deflection → steady-state rate = .af MAX\*\_RATE
- RATE\_KP = 0.7 / MAX\*\_RATE starting point (tuned from here where criteria demand)
- See FW\_Control\_Authority\_Calibration.md.

#### 4. Roll Structural Gate (v2.1)

- Aileron-less RudderElevatorAF frames → roll non-scored (bank-and-yank model gap).

### Fixed Wing Results — old vs revised

Format: KP/KI for angle loop, RKp/RKd for rate loop. **Bold** = changed in this pass.

#### Roll

| Frame            | Old Angle KP/KI | Old RKp/RKd | New Angle KP/KI    | New RKp/RKd        |
|------------------|-----------------|-------------|--------------------|--------------------|
| SkySurfer_Blixer | 0.5/0           | 0.223/0.05  | <b>0.8/0</b>       | <b>0.357/0.07</b>  |
| Shadow           | 6/0             | 0.334/0.012 | <b>7.8/0</b>       | <b>0.535/0.017</b> |
| Dragon           | 3/0.05          | 0.334/0.012 | 3/0.05             | 0.334/0.012        |
| SmallSpoiler     | 0.75/0.1        | 0.891/0.004 | 0.75/0.1           | 0.891/0.004        |
| Elevon           | 0.25/5          | 0.223/0.05  | <b>0.676/13.52</b> | <b>0.913/0.137</b> |
| Delta            | 7/2             | 0.223/0.05  | <b>11.2/3.2</b>    | <b>0.357/0.07</b>  |
| Spoileron        | 0.25/5          | 0.223/0.05  | <b>0.52/10.4</b>   | <b>0.570/0.098</b> |
| Radian           | 4/0.2           | 0.223/0.005 | 4/0.2              | 0.223/0.005        |
| RudderElevator   | 4/0.2           | 0.223/0.005 | 4/0.2              | 0.223/0.005        |
| Arado_555        | 1.5/0           | 0.334/0.05  | 1.5/0              | 0.334/0.05         |
| Horten           | 3/0.05          | 0.334/0     | 3/0.05             | 0.334/0            |
| Phoenix          | 2/0.1           | 0.434/0     | 2/0.1              | 0.434/0            |

Radian/RudderElevator/Phoenix roll is **non-scored** (aileron-less). Roll rows retained for record.

#### Pitch

| Frame            | Old Angle KP/KI | Old RKp/RKd | New Angle KP/KI | New RKp/RKd        |
|------------------|-----------------|-------------|-----------------|--------------------|
| SkySurfer_Blixer | 1/3             | 0.178/0     | 1/3             | 0.178/0            |
| Shadow           | 1/2             | 0.334/0     | 1/2             | 0.334/0            |
| Dragon           | 1/2             | 0.334/0     | 1/2             | 0.334/0            |
| SmallSpoiler     | 1/0.1           | 0.891/0.004 | <b>4/0.2</b>    | <b>2.5/0.004</b>   |
| Elevon           | 10/0.05         | 0.668/0.006 | 10/0.05         | 0.668/0.006        |
| Delta            | 10/0.2          | 0.668/0.006 | 10/0.2          | 0.668/0.006        |
| Spoileron        | 5/0.2           | 0.668/0.005 | <b>8/0.32</b>   | <b>1.069/0.007</b> |
| Radian           | 7/0.1           | 0.668/0.005 | 7/0.1           | 0.668/0.005        |
| RudderElevator   | 8/0.1           | 0.668/0.005 | 8/0.1           | 0.668/0.005        |
| Arado_555        | 1/2             | 0.334/0     | 1/2             | 0.334/0            |
| Horten           | 1/2             | 0.334/0     | 1/2             | 0.334/0            |
| Phoenix          | 2/0.1           | 0.668/0     | <b>8/0.2</b>    | <b>1.0/0.004</b>   |

#### Yaw

| Frame            | Old Angle KP/KI | Old RKp/RKd  | New Angle KP/KI | New RKp/RKd  |
|------------------|-----------------|--------------|-----------------|--------------|
| SkySurfer_Blixer | 1/0.5           | 0.668/0.0011 | 1/0.5           | 0.668/0.0011 |

| Frame         | Old Angle KP/KI | Old RKp/RKd  | New Angle KP/KI  | New RKp/RKd         |
|---------------|-----------------|--------------|------------------|---------------------|
| Shadow        | 3/0.25          | 0.334/0.0011 | 3/0.25           | 0.334/0.0011        |
| Dragon        | 3/0.25          | 0.334/0.0011 | 3/0.25           | 0.334/0.0011        |
| Small Spoiler | 1/1.5           | 0.334/0.0011 | 1/1.5            | 0.334/0.0011        |
| Elevon        | 3/0.2           | 0.134/0.0015 | 3/0.2            | 0.134/0.0015        |
| Delta         | 4/0.2           | 0.7/0.0015   | 4/0.2            | 0.7/0.0015          |
| Spoileron     | 8/0.3           | 0.134/0      | <b>10.4/0.39</b> | <b>0.214/0</b>      |
| Radian        | 10/0.05         | 0.134/0      | <b>13/0.065</b>  | <b>0.214/0</b>      |
| Rudder Elevon | 10/0.05         | 0.134/0      | <b>13/0.065</b>  | <b>0.214/0</b>      |
| Arado_555     | 3/0.25          | 0.334/0.0011 | 3/0.25           | 0.334/0.0011        |
| Horten        | 3/0.25          | 0.334/0.0011 | 3/0.25           | 0.334/0.0011        |
| Phoenix       | 3/0.25          | 0.111/0.0003 | <b>7.68/0.64</b> | <b>0.285/0.0007</b> |

Elevon/Shadow/Dragon/Arado\_555/Horten yaw is **non-scored** (rudderless elevon).  
Rows retained for record.

### Altitude Hold & Navigation (shared across FW frames)

| Param             | Value |
|-------------------|-------|
| ALT_POS_KP        | 0.4   |
| ALT_POS_KI        | 0.005 |
| ALT_ROC_KP        | 0.2   |
| UNUSED_ALT_VEL_KI | 0.01  |
| NAV_POS_KP        | 0.9   |
| NAV_POS_KI        | 0.005 |
| NAV_VEL_KP        | 2.4   |

(AH/Nav gains unchanged from v1.0 pass — all FW AH/Nav already passed.)

### Multicopter — Generic (5 frames)

| Frame       | AH                                           | Nav                          | Attitude     |
|-------------|----------------------------------------------|------------------------------|--------------|
| Quad        | KP=1.83<br>KI=0.005<br>RKP=0.2<br>RKI=0.01   | PK=1.5<br>KI=0.005<br>VK=0.2 | (stock PASS) |
| Quad_Medium | KP=1.83<br>KI=0.005<br>RKP=0.2<br>RKI=0.0005 | PK=1.5<br>KI=0.005<br>VK=0.2 | (stock PASS) |
| Quad_Racer  | KP=1.5<br>KI=0.005<br>RKP=0.2<br>RKI=0.0005  | PK=1.5<br>KI=0.005<br>VK=0.2 | (stock PASS) |

| Frame | AH                                           | Nav                          | Attitude                                               |
|-------|----------------------------------------------|------------------------------|--------------------------------------------------------|
| Hex   | KP=1.83<br>KI=0.005<br>RKP=0.2<br>RKI=0.0005 | PK=1.5<br>KI=0.005<br>VK=0.2 | (stock PASS)                                           |
| Oct   | KP=1.5<br>KI=0.005<br>RKP=0.2<br>RKI=0.0005  | PK=1.5<br>KI=0.005<br>VK=0.2 | <b>Roll KP=0.25, Pitch KP=0.20</b><br>(gust rejection) |

All 5 MR frames PASS (re-verified 2026-08-04 — unaffected by FW sim changes).

## Original Frames (Critique → Retune)

original/Phoenix.af was critique-only at v1.0. **Retuned in this pass** — now **PASS** (pitch/yaw tuned up; roll non-scored). See `critique_report.md` for the pre-retune baseline of all original frames.

## Structural Notes (Not Tuning Failures)

1. **Aileron-less RudderElevatorAF** (Radian, RudderElevator, Phoenix): roll authority is zero in the coupled model (bank-and-yank cancelled by yaw hold) → roll non-scored (v2.1 gate).
2. **Rudderless elevons** (Dragon, Elevon, Shadow, Arado\_555, Horten): yaw authority holds only ~1-3° against weathervane → yaw non-scored.
3. **SmallSpoileron pitch**: low pitch authority (CM\_D\_ELE 0.45, strong static margin −0.255) — needed ~3× rate KP (2.5 vs anchor 0.891) to reach the 30° step. This is at the physical limit; higher rate KP starts to reduce disturbance margin.
4. **Shadow free-flight roll** (~1286° divergence under turbulence): intrinsic low-dihedral (0.05) flying-wing characteristic — open-loop, not a PID failure.

## Parameter System Notes

- All 128 params stored as float32 in FC (`Config.ParamData[i].f`)
- Display: `display = fc_float * PARAM_DISPLAY_MULT[idx]`
- Write: `fc_float = display / PARAM_DISPLAY_MULT[idx]`
- Only selector/enum params remain u8 (ArmingMode, RxType, ConfigBits, etc.)
- No legacy uint8 path, no legacy checkbox

## Files Modified

### FC Firmware (UAVXArmQ)

- `src/params.c` — ParamTable outer-loop bounds (7 tags)

## GCS (UAVXGS)

- uavx-python/src/parameters.py — PARAM\_LIMITS mirrored for 7 tags
- uavx-python/src/tests/test\_pid\_sim.py — AH\_CRITERIA\_FW relaxed (6s/8s); **v2.0** authority anchor + get\_fw\_descriptor/\_compute\_fw\_max\_rates; **v2.1** is\_roll\_structurally\_limited + Roll gating
- /tmp/opencode/auto\_tune.py — coordinate-search tuner (rebuilt this pass)

## Airframes (.af files)

All 12 FW .af files re-tuned (see tables above); 5 MR .af unchanged.

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## Next Steps (Tracked in TODO)

1. **RudderMotorFF** — New param for differential motor thrust (even-motor-count FW)
  2. **AH Tuning** — Overshoot on descent (FW)
  3. **Nav Gain Tuning** — Nav.PosKp, Nav.PosKi, Nav.MaxVelocity, Nav.MaxBankAngle
  4. **Yaw Rate Integral** — Add I term in rate loop for heading hold
  5. **Bank-and-yank model** — Proper RudderElevatorAF roll via rudder→yaw→dihedral coupling (removes the roll structural gate)
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## Verification

Run full critique on any frame:

```
python3 -m test_pid_sim generic/Dragon.af      # PASS
python3 -m test_pid_sim generic/Delta.af       # PASS
python3 -m test_pid_sim user/Arado_555.af      # PASS
python3 -m test_pid_sim original/Phoenix.af    # PASS
python3 -m test_pid_sim generic/Quad.af        # PASS (MR)
```

**2026-08-04 v2.1 verification: all 12 FW + 5 MR frames PASS.**